The Central Limit Theorem

Statistics 12 — A 90-Minute Lesson

Lesson Goals

By the end of this lesson, students should be able to:

- explain what a **sampling distribution** is
- state the Central Limit Theorem (CLT) precisely
- identify the **mean** and **standard error** of a sampling distribution
- recognise when the CLT applies
- use the CLT to calculate probabilities about sample means
- distinguish between the distribution of *individuals* and the distribution of *sample means*

Suggested Timing

Time	Activity
0–5 min	Hook: the dice simulation
5–20 min	Sampling distributions — concept & vocabulary
20–35 min	The CLT statement & parameters
35–50 min	Worked examples
50–60 min	Think-pair-share & misconceptions
60–75 min	More worked examples (harder)
75–85 min	Student practice
85–90 min	Summary & exit card

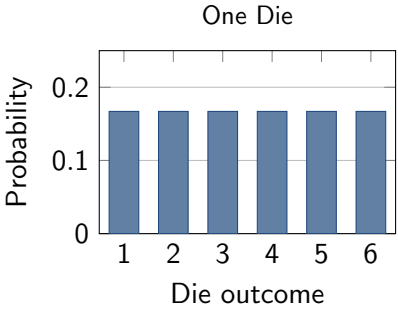
Hook: Rolling Dice

Imagine rolling **one fair die** many times.

- Each outcome (1–6) is equally likely.
- The distribution is **uniform** — definitely not bell-shaped.

Now imagine rolling **30 dice** and recording the *average*.

- Repeat this experiment thousands of times.
- What shape do you expect for the distribution of averages?



The surprising answer

The distribution of sample averages will look approximately **normal**, no matter the shape of the original population!

Key Vocabulary

Population

The complete set of individuals or measurements we are interested in.

Sample

A subset of the population that we actually observe.

Statistic

A number computed from a sample (e.g. the sample mean $\bar{x}$).

Sampling Distribution

The distribution of a statistic over *all possible samples of the same size* drawn from the same population.

Key insight: because samples vary, statistics vary. A sampling distribution describes *how much* and *in what pattern*.

Population vs. Sampling Distribution

Distribution of individuals

- Describes one observation chosen at random
- Mean: μ
- Standard deviation: σ
- Can be any shape

Sampling distribution of $\bar{x}$

- Describes the mean of a sample of size n
- Mean: $\mu_{\bar{x}} = \mu$
- Standard deviation: $\sigma_{\bar{x}} = \dfrac{\sigma}{\sqrt{n}}$
- Approximately normal (CLT!)

Standard Error

The quantity $\dfrac{\sigma}{\sqrt{n}}$ is called the **standard error** (SE) of the mean. It measures how much $\bar{x}$ typically varies from sample to sample.

The Central Limit Theorem — Statement

Central Limit Theorem (CLT)

Let $X_1, X_2, \ldots, X_n$ be a random sample from *any* population with mean μ and finite standard deviation σ. When n is sufficiently large, the sampling distribution of the sample mean $\bar{X}$ is approximately normal:

$$\bar{X} \approx N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

The approximation improves as n increases.

Three things the CLT tells us about $\bar{X}$:

1. **Shape:** approximately normal
2. **Centre:** mean equals the population mean μ
3. **Spread:** standard deviation equals $\dfrac{\sigma}{\sqrt{n}}$

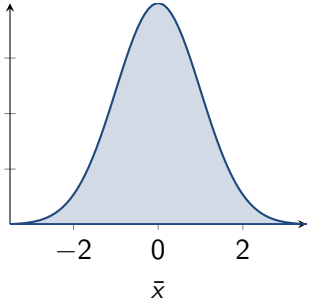
When Does the CLT Apply?

The usual guideline

$n \geq 30$ is sufficient for most populations.

- If the population is already **normal**, the CLT holds for *any* n.
- For **mildly skewed** populations, $n \approx 15$–20 may suffice.
- For **heavily skewed** or unusual populations, you may need $n > 30$.

Sampling distribution of $\bar{X}$ (large n)



Important caveat

The CLT applies to the distribution of *sample means*, not to individual values.

Why the Standard Error Shrinks

The standard error $\frac{\sigma}{\sqrt{n}}$ decreases as n grows.

n	$\sigma = 10$	SE $= \sigma/\sqrt{n}$	Relative to $n = 1$
1	10	10.00	100%
4	10	5.00	50%
25	10	2.00	20%
100	10	1.00	10%
400	10	0.50	5%

- To **halve** the standard error, you must **quadruple** the sample size.
- Larger samples give more precise estimates of μ.

Using the CLT — The Z-Score for $\bar{X}$

Standardising a sample mean

To find probabilities about $\bar{X}$, convert to a z-score using:

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

Then use a standard normal table or calculator.

Compare:

- For an individual: $z = \dfrac{x - \mu}{\sigma}$
- For a sample mean: $z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}}$

The only difference is the **denominator** — we divide by the standard error, not σ.

Worked Example 1: Coffee Shop Wait Times

Problem

Wait times at a coffee shop are right-skewed with mean $\mu = 4.2$ min and standard deviation $\sigma = 1.8$ min. A random sample of $n = 36$ customers is selected. What is the probability that the sample mean wait time exceeds 4.8 minutes?

Step 1: Check CLT conditions. $n = 36 \geq 30$. ✓

Step 2: Identify the sampling distribution.

$$\bar{X} \approx N\left(4.2,\ \frac{1.8}{\sqrt{36}}\right) = N(4.2,\ 0.3)$$

Step 3: Standardise.

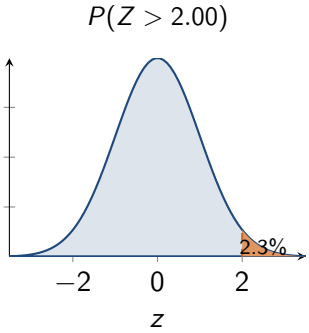
$$z = \frac{4.8 - 4.2}{0.3} = \frac{0.6}{0.3} = 2.00$$

Step 4: Find the probability.

$$P(\bar{X} > 4.8) = P(Z > 2.00) = 1 - 0.9772 = \mathbf{0.0228}$$

Worked Example 1: What Does This Mean?

- There is about a **2.3% chance** that a sample of 36 customers has a mean wait time above 4.8 minutes.
- This is quite unlikely — a sample mean that far above μ would be unusual.
- Notice: for an *individual*, $P(X > 4.8)$ would be much larger.



Key point

Sample means are less variable than individual values.

Worked Example 2: Heights of Students

Problem

Heights of male students are normally distributed with $\mu = 175$ cm and $\sigma = 8$ cm. A random sample of $n = 16$ male students is selected.

(a) Find $P(\bar{X} < 172)$.
(b) Find $P(170 < \bar{X} < 178)$.

Since the population is normal, CLT holds for any n.

$$\bar{X} \sim N\left(175,\ \frac{8}{\sqrt{16}}\right) = N(175,\ 2)$$

Part (a):

$$z = \frac{172 - 175}{2} = -1.5 \qquad P(\bar{X} < 172) = P(Z < -1.5) = \mathbf{0.0668}$$

Part (b):

$$z_1 = \frac{170 - 175}{2} = -2.5 \qquad z_2 = \frac{178 - 175}{2} = 1.5$$

Think-Pair-Share: Spot the Mistake

A student wrote the following. What is wrong?

"The population of exam scores has mean 70 and standard deviation 15. A sample of $n = 25$ students is taken. The probability that a student scores above 75 is $P(Z > 1.33) \approx 0.092$. Therefore, about 9% of sample means will be above 75."

Discuss with a partner, then share your answer.

Hint

The student computed $z = \dfrac{75 - 70}{15} = 1.33$. Is this the right formula for a sample mean?

Think-Pair-Share: Correction

What went wrong

The student used $z = \dfrac{\bar{x} - \mu}{\sigma}$ instead of $z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}}$.

The correct calculation:

$$\text{SE} = \frac{15}{\sqrt{25}} = 3 \qquad z = \frac{75 - 70}{3} = 1.67$$

$$P(\bar{X} > 75) = P(Z > 1.67) \approx 0.0475$$

The probability is about **4.8%**, not 9.2%.

Common error to avoid

When working with sample means, *always* divide by $\dfrac{\sigma}{\sqrt{n}}$, not by σ alone.

Worked Example 3: Finding a Boundary Value

Problem

Monthly electricity bills in a city have mean \$120 and standard deviation \$25. A random sample of $n = 100$ bills is selected. Find the value c such that $P(\bar{X} > c) = 0.05$.

Step 1: Sampling distribution.

$$\bar{X} \approx N\left(120,\ \frac{25}{\sqrt{100}}\right) = N(120,\ 2.5)$$

Step 2: Find the z value. We need $P(Z > z^*) = 0.05$, so $z^* = 1.645$.
Step 3: Unstandardise.

$$c = \mu + z^* \cdot \frac{\sigma}{\sqrt{n}} = 120 + 1.645 \times 2.5 = 120 + 4.11 = \mathbf{\$124.11}$$

Only 5% of samples of size 100 would have a mean bill above \$124.11.

Worked Example 4: Two Populations

Problem

Brand A batteries have mean life $\mu_A = 500$ h, $\sigma_A = 40$ h.
Brand B batteries have mean life $\mu_B = 490$ h, $\sigma_B = 30$ h.
Samples of $n = 64$ batteries from each brand are tested independently. What is the probability that Brand A's sample mean exceeds Brand B's by more than 20 hours?

Let $D = \bar{X}_A - \bar{X}_B$. By the CLT:

$$E[D] = 500 - 490 = 10 \qquad \mathrm{SD}(D) = \sqrt{\frac{40^2}{64} + \frac{30^2}{64}} = \sqrt{25 + 14.06} = \sqrt{39.06} \approx 6.25$$

$$z = \frac{20 - 10}{6.25} = 1.60 \qquad P(D > 20) = P(Z > 1.60) \approx \mathbf{0.0548}$$

Student Practice

Work on these problems individually or with a partner.

1. The weights of apples at a farm are distributed with $\mu = 182$ g and $\sigma = 14$ g. A crate contains $n = 49$ apples chosen at random.
   a. Find the mean and standard error of $\bar{X}$.
   b. Find $P(\bar{X} > 185)$.
   c. Find $P(179 < \bar{X} < 186)$.
2. A random sample of $n = 64$ light bulbs is drawn from a population with $\mu = 800$ h and $\sigma = 80$ h. Find the value c such that $P(\bar{X} < c) = 0.90$.
3. A population is strongly right-skewed with $\mu = 50$ and $\sigma = 10$. Can you apply the CLT to a sample of $n = 10$? Explain your reasoning.

Practice Solutions

1. (a) $\mu_{\bar{X}} = 182$ g, SE $= 14/\sqrt{49} = 2$ g.

(b) $z = (185 - 182)/2 = 1.5 \Rightarrow P(\bar{X} > 185) = P(Z > 1.5) = 0.0668$.

(c) $z_1 = (179 - 182)/2 = -1.5$, $z_2 = (186 - 182)/2 = 2.0$
$P(-1.5 < Z < 2.0) = 0.9772 - 0.0668 = 0.9104$.

2. SE $= 80/\sqrt{64} = 10$. Need $P(Z < z^*) = 0.90 \Rightarrow z^* = 1.282$.
$c = 800 + 1.282 \times 10 =$ **812.8** h.

3. No. The guideline requires $n \geq 30$ for non-normal populations, and strongly skewed populations may need $n > 30$. With $n = 10$, the CLT cannot reliably be applied here.

Common Misconceptions

Misconception	Correction
"The CLT says the population becomes normal."	The CLT is about the *sampling distribution of* $\bar{X}$, not the population.
"A larger n makes individual values less variable."	n only reduces the variability of *sample means*, not individuals.
"CLT requires the population to be symmetric."	The CLT works for any shape, as long as n is large enough.
"$\sigma/\sqrt{n}$ is the standard deviation of the population."	It is the *standard error* of $\bar{X}$, not σ.

Lesson Summary

The Central Limit Theorem

For a random sample of size n from a population with mean μ and standard deviation σ:

$$\bar{X} \approx N\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \quad \text{when } n \geq 30 \text{ (or population is normal)}$$

Key take-aways:

- The **mean** of $\bar{X}$ equals the population mean μ.
- The **standard error** is $\sigma/\sqrt{n}$ — it shrinks with larger n.
- Use $z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ to find probabilities.
- The CLT applies regardless of population shape (for large n).
- The CLT is the foundation for **confidence intervals** and **hypothesis tests** (coming soon!).

Exit Card

Answer on a piece of paper before you leave.

Question

The time (in minutes) students spend on homework each night has mean $\mu = 45$ min and standard deviation $\sigma = 12$ min. A random sample of $n = 36$ students is selected.

a. State the sampling distribution of $\bar{X}$, including its shape, mean, and standard deviation.
b. Find $P(\bar{X} > 48)$.

Answer

(a) $\bar{X} \approx N(45,\ 2)$ since $12/\sqrt{36} = 2$. (b) $z = (48 - 45)/2 = 1.5$; $P(Z > 1.5) = 0.0668$.